

Philosophical Logic Task E

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March 9, 2025

Discuss whether we should except the following (500 words each):

(a) If $\Gamma \models C$, then $\{A \Box \rightarrow B : B \in \Gamma\} \models A \Box \rightarrow C$.

Response: The statement, if $\Gamma \models C$, then $\{A \Box \rightarrow B : B \in \Gamma\} \models A \Box \rightarrow C$, reads if a set of wffs semantically entail a conclusion C , and if A obtains, then every element in Γ would obtain, then it follows that if A obtains, C would obtain. According to Stalnaker in *Inquiry* (1984), this principle follows from accepting the limit assumption, namely that if some ϕ -world exists, then for every world $w \in \mathcal{W}$ there exists a maximally closest ϕ -world to w . There are two ways we can assess the principle. The first involves parsing through what the principle means, and assessing whether we should accept the limit assumption on that grounds. Let us call this the principle-first approach. The second involves directly challenging the limit assumption; let us call this the metaphysical approach. I am going to argue that both approaches coincide in that we should deny the principle.

The metaphysical approach is easier. Lewis gives the intuitively false statement 'If a line had been longer than one inch, it would be a 100 miles long'. The reason why it is true on the account of SC-semantics is because $V_{\mathcal{M}}(\phi \Box \rightarrow \psi, w) = 1$ iff $V_{\mathcal{M}}(\phi, u) = 0, \forall u \in \mathcal{W}$ or if $V_{\mathcal{M}}(\phi, v) = 1$ and for all $y \in \mathcal{W}$ s.t. $V_{\mathcal{M}}(\phi, y) = 1$ $v \leq_w y$, then $V_{\mathcal{M}}(\psi, v) = 1$. But there is at least one world in which $V_{\mathcal{M}}(\phi, u) = 1$ but there is no such closest ϕ -world due to the density of the rationals in the reals (so for any number $x \in \mathbb{R}$ you pick, there are infinite $q \in \mathbb{Q}$ in between 1 and x). I think that this line of argument could be made more direct – what world could be maximally closest to w except for the world w itself (if ϕ was true in w)? When we think of metaphysical possibility, we think of it as possibility most broadly construed – this means that there plausibly is a continuum of possible ϕ -worlds in between any ϕ -world and w , because there are just so many possibilities. From a metaphysical standpoint, the limit assumption is clearly intolerable. So on this account, we should reject the principle in (a).

Now, we can take the opposite approach to Lewis by considering what the principle in (a) says. Basically, we need to consider if we can accept the following: suppose if some state of affairs, A , would happen, then a set of events would happen, call this $\Gamma = \{B_1, \dots, B_n\}$. Now, collectively, $\Gamma \models C$. This principle seems compelling: if A had happened, all of Γ members would obtain in some closest A -world, and since B semantically entails C , C must obtain in that closest A -world. The assumption here seems to also be one of anti-symmetry (which I will reject later on). Once we lose anti-symmetry, we realise that if there are multiple closest A -worlds, then it is not true that all of Γ members have to obtain together in the same world, and hence C can be false in any of these closest worlds. Now, Stalnaker seems to suggest that (a) is independent of anti-symmetry and I cannot see why, but nonetheless, I have given us reason to reject the principle as is.

(b) $\models (A \Box \rightarrow C) \vee (A \Box \rightarrow \sim C)$.

Response: The principle presented $\models (A \Box \rightarrow C) \vee (A \Box \rightarrow \sim C)$ is the conditional law of the excluded middle: namely, if A had happened, then C would happen or if A had happened, then $\sim C$ would happen. The reason for this, according to Lewis, is because of the principle of anti-symmetry. I will consider two approaches again – the first challenging the principle itself and the second examining the metaphysics behind the principle.

Let us first consider the principle itself. Lewis suggests that the principle at first pass seems compelling, but licences flawed inferences. Consider: if I had flipped a penny, it would land on heads or if I had flipped

a penny, it would land on tails. The flaw in this is that, while it is true that if I had flipped a penny, it would land on heads or tails, it is only true that if I had flipped a penny, it *might* land on heads or if I had flipped a penny, it *might* land on tails. Notice that *would* implies a sense of necessity: if $A \Box \rightarrow B$, and if A obtained in some closest world to w , then B must obtain. But it is not true that a coin must land on tails if I flipped it in some closest world w and neither is it true that a coin must land on tails if I flipped it in some closest world w . So, while this principle is initially plausible, once we realise the distinction between might and would, the reason for denying it on the grounds of the principle itself becomes clear.

Now, we consider the metaphysics behind the antisymmetry assumption. Intuitively, as Lewis suggests, the similarity relation should not be anti-symmetric – two distinct cars can be as similar to another car, but on different accounts. Stalnaker suggests that this intuition stems, not from the fact that there can be actual ties in our similarity relation, but from the vagueness of our semantics and natural language – similarity is fundamentally vague and so we cannot parse through two different accounts of differences and assess whether they really are equivalent. In turn, he suggests a supervaluationist approach – if on the account of all these vaguely equally similar worlds, ψ is false or true, then it does not matter. Two remarks: first, Stalnaker does not make it clear how his theory would accommodate indeterminacy which can result from precisification; second, I disagree that we think that there are ties only as a feature of the vagueness of our natural language. It can be admitted that there is vagueness in notions of similarity, but this is not ipso facto a reason to think that two distinct objects can't be equally similar to some other object. For instance, consider some interval $(-5, 5)$. The intervals $(-5, 10)$ and $(-10, 5)$ are clearly distinct but equally close. But there is a separate, and perhaps stronger response is Lewis's observation that with the anti-symmetry assumption, we do not distinguish between the might and would counterfactual conditions since under SC, $\phi \Diamond \rightarrow \psi$ implies $\phi \Box \rightarrow \psi$ and obviously, $\phi \Box \rightarrow \psi$ implies $\phi \Diamond \rightarrow \psi$. But might and would are obviously very different conditions – so we have reasons to reject the anti-symmetry assumption and the law of the excluded middle.